# Education as the Sole Primary Driver of Human Development: Against Sen's Bifurcation of Development Pathways

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## Abstract

Amartya Sen's distinction between growth-mediated and support-led security has structured development policy debates for four decades. We argue the dichotomy is not merely incomplete —it is wrong in its identification of cause. Neither economic growth nor direct state provision is an independent mechanism of human development. Income is a downstream consequence of education. Direct provision is infrastructure that educated populations demand through political pressure and utilize through micro-decisions —it works when education is present to activate it, and stalls or reverses when provision is removed from an educated population, as China's post-1980 health dismantling shows. Education is the necessary and primary driver; everything else is contingent on it.

We define development concretely as crossing both thresholds simultaneously: total fertility below the United States in 1960 (TFR < 3.67) and life expectancy above it (LE > 70.1 years) —two outcomes that education directly drives and that do not require high income to achieve. Bangladesh's TFR crossed in 2000 but development is recorded at 2011, when the LE threshold was also crossed. Using 189-country panel data (1975–2015) from the Wittgenstein Centre for Demography and Global Human Capital (WCDE v3) with country fixed effects and a 25-year generational lag, we show that parental lower secondary completion predicts within-country educational outcomes at $\beta=0.485$ ($R^2=0.464$), nearly double the explanatory power of GDP alone ($R^2=0.266$). A century-long panel (28 countries, 1900–2015) finds near-unity generational transmission (FE $\beta=0.960$). Education at T predicts GDP, life expectancy, and total fertility 25 years forward after controlling for initial conditions, confirming education's upstream causal position.

Sen's canonical support-led cases —Kerala, China, and Sri Lanka — each crossed the development threshold at precisely the interval that PTE (Parental Transmission of Education, ~25 years per generation) predicts from their prior education investments: Kerala ~1982, Sri

Lanka 1991, China 1997. Taiwan crossed in 1972, South Korea in 1986 —the same mechanism expressing itself differently. The variation in speed is explained by the rupture margin (how decisively the expansion exceeded the base-adjusted threshold rate) and whether structural disruption to the LE pathway occurred during the compounding period. The generational transmission mechanism operates below the level at which political disruptions occur. PTE does not require states to propagate once an educated population exists — it requires parents. State disruption and civil war affect institutions and economies; they do not interrupt the household transmission of educational aspiration from parents to children. What states are required for is the prior step: initiating and sustaining the educational investment that creates the educated population in the first place.

These findings extend Easterlin (1981) and Lutz & Kebede (2018) to the within-country identification of the generational mechanism and directly challenge institutions-first accounts: the parental education coefficient survives country fixed effects absorbing all time-invariant institutional quality.

**Keywords:** education; generational transmission; human development; Preston Curve; PTE; GPTE; GGPTE; Sen; leapfrog development; educational rupture

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## 1. Introduction

In *Hunger and Public Action* (Drèze & Sen, 1989), Amartya Sen drew a distinction that became foundational to development thinking: some countries secure welfare through economic growth (growth-mediated security); others through direct state provision regardless of income (support-led security). Kerala, Sri Lanka, and China were the paradigm cases of the second type —populations with welfare outcomes far above what their incomes predicted.

The framework was politically important. It challenged the then-dominant view that growth was the necessary precondition for welfare and opened space for direct intervention. Its influence on the Millennium Development Goals, the Human Development Index, and direct service delivery programmes across the developing world is difficult to overstate.

We argue the framework is wrong in a specific and consequential way —not incomplete, but causally backwards on both ends. Sen's two categories are not competing mechanisms. They are the same mechanism —education —producing two different surface appearances depending on the speed of educational expansion and the presence of

functioning markets. Where educational expansion is fast and markets function, the mechanism produces rapid visible income growth; that income growth gets credited as the cause. Where expansion is slow and markets are absent, income growth is invisible; state services are the only observable welfare activity and they get credited instead. Sen observed the surfaces and built a causal typology from them.

Education is the sole primary driver. This claim requires precise definition. It does not mean education explains all variance in welfare outcomes, or that income and institutions have zero effect at the margin. It means education is the *necessary condition* —the asymmetric input without which no other mechanism operates independently. Remove income, and education still works: Bangladesh crossed both development thresholds at $1,250 per capita. Remove the state health apparatus, and education still works: Korea and Taiwan achieved development with minimal welfare states. Remove education, and neither income nor services work: Uganda at $1,000 per capita had higher life expectancy than India in 1960, then watched Amin destroy both the provision and the educated population that could have rebuilt it. The asymmetry defines primacy. No other variable survives this test.

The institutional consequence is direct. OECD Development Assistance Committee data shows health has consistently received 2–3× the aid allocation of education across the post-MDG period; within education aid, secondary receives a fraction of what primary receives (OECD DAC, 2022). Every low-income finance minister knows that announcing a health programme attracts bilateral donors and multilateral loans; announcing a secondary school construction programme attracts far less. This tilt is the policy legacy of Sen' s framework and it systematically underfunds the only mechanism that compounds across generations.

The Millennium Development Goals organized reporting around five direct health metrics and one primary enrollment target —not secondary completion. The SDGs made the hierarchy explicit: SDG 1 is No Poverty, SDG 2 is Zero Hunger, SDG 3 is Good Health, SDG 4 is education. This ordering is not an accident of numbering; it is Sen' s causal model institutionalized. The MDG/SDG health achievements —child mortality reductions, maternal mortality improvements —are not evidence that this ordering is correct. They are evidence that the global educational rupture since 1950 produced the institutional capacity to deliver health technology internationally: WHO, UNICEF, GAVI, and PEPFAR are staffed by educated workers deploying welfare technology to populations still building their own educational base. Those gains are real and do not compound. They persist only

as long as external delivery continues and do not change the underlying generational trajectory unless domestic educational investment follows.

The purpose of this paper is to establish the education-only account empirically and theoretically. Section 2 develops the theoretical framework from Easterlin (1981) through Lutz & Kebede (2018) to the generational transmission mechanism (PTE). Section 3 defines development concretely. Section 4 addresses causal identification —the bad control problem and natural experiments. Section 5 presents the empirical evidence. Section 6 reports results. Section 7 provides the decisive test: when did Sen's canonical cases actually develop, and does the timing track education or tempo of expansion? Section 8 addresses the institutional challenge. Section 9 discusses the educational rupture as the one exogenous political decision from which everything else follows. Section 10 states limitations.

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## 2. Theoretical Framework

### 2.1 Easterlin's Founding Argument

The education-first account of development has a precise starting point. Easterlin (1981), in "Why Isn't the Whole World Developed?", asked why modern economic growth had failed to diffuse uniformly given the post-war availability of production technology. His answer was specific: the binding constraint was the absence of mass schooling systems. Technology transfers when it encounters a literate, numerate population capable of deploying it. Without that population, it does not transfer, regardless of capital access or external assistance.

Easterlin located this empirically in the historical spread of education systems. The divergence in economic growth between early and late industrializers tracked the divergence in mass schooling —driven originally by the Protestant Reformation's insistence on personal scripture-reading, which produced mass literacy as a side-effect of religious practice across Northern Europe. The distribution of development in 1960 was substantially the distribution of schooling in 1860.

### 2.2 Lutz and the Preston Curve

Lutz & Kebede (2018) take Easterlin's historical argument into the data. They reinterpret the Preston Curve (Preston 1975) —the positive relationship between national income and life expectancy —by re-

placing income with education on the x-axis. Three findings emerge. First, the education–life expectancy relationship is tighter than the income–life expectancy relationship. Second, and more important, the upward shift of the curve over time —the fact that populations live longer today at the same income levels than in 1960 —largely disappears when education is on the x-axis. The "wealth buys health" reading of the Preston Curve is an education story that income was proxying for, because income and education are correlated but income is downstream of education in the causal chain (Lutz 2009). Third, and directly relevant to the scope of educational investment: the education–life expectancy relationship shows no diminishing returns at higher education levels. Unlike income, which follows the classic concave Preston shape, the education curve remains approximately linear from primary through tertiary. Lutz attributes this to a cognitive mechanism: education changes the brain's capacity for planning, learning from mistakes, and health-oriented decision-making across the entire life course, and these gains accumulate at all levels of schooling. The implication is that the case for educational investment does not weaken as countries progress toward universal secondary; pushing further into upper secondary and tertiary produces continued life expectancy gains. This is consistent with our WCDE panel: among 72 countries with lower-secondary completion above 85% in 2010, college completion carries a within-group correlation of r=0.51 with life expectancy, with mean life expectancy rising from 73.5 years in the lowest college-completion quartile to 80.3 years in the highest —a nearly 7-year gradient that emerges among countries that have already solved the lower-secondary problem.

Deaton (2013) offers an alternative account of the curve's upward shift: global diffusion of health knowledge and technology —antibiotics, vaccines, oral rehydration therapy —produced life expectancy gains independent of both income and education. This account is not inconsistent with the education-primary claim but clarifies the mechanism: technology diffuses in principle to all countries simultaneously, yet uptake is not uniform. The differential uptake tracks education. It is educated mothers who consistently vaccinate their children; it is literate communities that follow oral rehydration protocols; it is educated populations that build the political demand for the health infrastructure that distributes the technology. Deaton identifies what became available globally; education identifies who could use it.

Lutz & Kebede (2018) document one further finding that bears directly on this point: life expectancy at the lowest education level in 2010 *exceeds* the 1970 baseline for that same level —the only level at which a genuine upward Preston Curve shift remains when education replaces income on the x-axis. This residual is the global

educational rupture operating through international institutions: educated actors in developed countries produced vaccines, antibiotics, and oral rehydration protocols; educated supply chains distributed them; educated administrators at WHO and UNICEF directed them to the populations with the lowest educational bases. This is provision —but provision that is education-mediated at the system level. It is the global analogue of the barefoot doctor argument: the doctors arrived from outside because the educational base inside was absent. Sen's "support-led security" in its MDG-era form is not evidence that provision works without education. It is evidence of the global educated class —concentrated in wealthy countries and international institutions —functioning as an external educated parent for populations that had not yet built their own educational base. The mechanism is still PTE; the scale is international rather than household.

### 2.3 The Generational Transmission Mechanism: PTE, GPTE, GGPTE

The micro-level process connecting these macro observations is generational transmission. We introduce a three-level taxonomy that unifies the observed variation in development lag lengths:

- **PTE** (Parental Transmission of Education): the child's educational attainment is predicted by the parent's, with an approximately 25-year lag. This is the base unit of the mechanism.
- **GPTE** (Grandparental Transmission of Education): the grandparent's education predicts the grandchild's, operating across approximately 50 years through two sequential PTE cycles.
- **GGPTE** (Great-Grandparental Transmission of Education): three cycles, approximately 75 years.

The unit interval is not assumed. A woman completing lower secondary at age 15–18 has children reaching school age 20–30 years later. Across a 140-year panel (28 countries, 1900–2015), the within-country PTE transmission coefficient approaches unity (FE $\beta$=0.960): a 1 percentage point gain in one generation predicts nearly a 1 percentage point gain in the next, compounding forward. The full lag profile —measured empirically at every year from lag 0 to lag 100 — confirms that the transmission signal persists across all three generational depths: $R^2$=0.369 at lag 25 (PTE), $R^2$=0.194 at lag 50 (GPTE), $R^2$=0.112 at lag 75 (GGPTE) (Figure A1; outcome: life expectancy, within-country FE).

The taxonomy directly explains the variation in development lag lengths across cases. Countries whose educational rupture was decisive and recent crossed at PTE distance. Countries whose ed-

ucational base was built gradually across two generations crossed at GPTE distance. Countries where education accumulated through social reform over three or more generations crossed at GGPTE distance. The lag length is not a free parameter fitted to match each case —it is the number of generational cycles required to compound from the starting educational base to the level at which the demographic transition engages. This answers directly why Taiwan crossed at ~20 years and Kerala at ~60–70 years: they are PTE and GGPTE cases respectively, determined by when their educational ruptures occurred and from what base.

The compounding has a structure that varies with the level of completion. Fertility declines continuously as female secondary completion rises; the steepest demographic headwinds are at the lowest completion levels, where total fertility rates are typically above 4, population growth is rapid, and the gains from each educated cohort are diluted across a larger and faster-growing next generation. PTE still operates and the system still advances at any level —but more slowly at low bases, against the headwind of high fertility and thin household resources. Korea' s data shows steady gains throughout the range from 25% to 95%, with no inflection point at which the demographic conditions change abruptly. Sustained state investment is required all the way to near-universal completion.

Critically, PTE/GPTE/GGPTE are household mechanisms. They operate through millions of individual family decisions —whether to send a daughter to secondary school, how many children to have, whether to vaccinate, how to interpret health information. States can create conditions for these decisions; they cannot make them. Political upheaval and civil war are simply irrelevant to the generational transmission mechanism, because they operate at a different level of society. The mechanism does not need to survive disruption —it runs beneath it.

The PTE taxonomy names the nuclear household channel but does not exhaust the transmission network. Two additional channels operate at shorter timescales and explain why aggregate educational gains can sometimes move faster than a strict 25-year reading of PTE predicts.

The **sibling channel** operates within the same household at variable lags: 2–20 years, depending on family size. In high-fertility societies, the gap between any two siblings can range from 2 to 20 years, so across a full sibship the oldest-to-youngest span can approach PTE timescales within the same family; an oldest sibling completing lower secondary transmits educational norms, tutoring, and aspirational expectations to the youngest siblings at lags approaching the parental interval. As fertility falls —itself a consequence of rising fe-

male education —sibling gaps narrow, sibling cohorts shrink, and this within-family channel weakens. The mechanism is self-limiting in exactly the right direction: sibling transmission is strongest when it is most needed (early educational transition, high fertility, low parental base) and attenuates as PTE from educated parents takes over. This also explains the smooth decay in Figure A1. If PTE were a single discrete 25-year mechanism, the $R^2$ lag profile would show a peak at lag 25 and rapid decay on either side. Instead, the profile decays gradually across lags 0–100. The sibling channel contributes signal at lags 5–20 years; PTE at lag 25; GPTE at lag 50; GGPTE at lag 75. The smooth aggregate curve is the empirical signature of these overlapping channels —a theoretical prediction of the multi-channel structure, not a statistical artifact or evidence against specificity.

The **extended family channel** operates across aunts, uncles, and grandparents present in child-rearing (2–25 years). This channel is disproportionately strong in extended-family household structures —South Asia, sub-Saharan Africa, East Asia —precisely the societies where development intervention matters most and where intergenerational household composition is densest. An educated aunt or uncle in the household independently predicts niece and nephew attainment above what parental education alone forecasts. The policy implication is not purely academic: educating any member of an extended family household produces downstream effects beyond that individual's own future children. The return on educating one woman ripples through the kin network at timescales shorter than the nuclear PTE interval.

The mechanism can also operate at supranational scale. Where a country lacks the domestic educational base to generate PTE-mediated welfare improvement from within, international institutions staffed by educated workers from educated countries can deliver welfare technology —vaccines, oral rehydration protocols, nutrition programmes —that produces welfare gains the domestic educational base could not yet sustain independently. This is what we term *global PTE*: the global educated class acting, collectively, as an educated parent for populations at low completion levels. It is still education-mediated; the education is simply located in the institutions delivering the intervention rather than in the households receiving it. The critical distinction is that global PTE produces welfare gains that are externally sustained —durable only as long as the international institutions maintain them —while domestic PTE produces gains that compound into the next generation. Global PTE buys time; domestic PTE changes trajectory.

### 2.4 Energy as Physical Substrate

The argument that education is the sole primary driver of human development must be distinguished from the claim that physical resources are irrelevant. They are not. Energy —the physical capacity of a civilization to do work —sets the ceiling on what any society can achieve materially. All economic transactions are, at their physical foundation, energy conversion events: manufacturing moves atoms, transport moves mass, health systems require cold chains and pharmaceutical supply chains, schools require heated buildings and printed materials. No economic activity occurs without energy input. This is thermodynamics, not metaphor.

The relationship between energy and education is hierarchical, not competitive. Energy is the substrate; education is the efficiency multiplier that determines how much of the energy ceiling a society actually reaches, and whether the gains compound into the next generation. The decisive evidence comes from opposite ends of the resource spectrum.

**Energy without education: the resource curse.** Venezuela holds the world's largest proven oil reserves. The Democratic Republic of Congo holds uranium, cobalt, coltan, and gold —inputs to the energy and technology supply chains of the developed world. Saudi Arabia, Kuwait, and Qatar have extracted trillion-dollar oil revenues over fifty years. None has crossed both development thresholds as defined in Section 3 through domestic compounding. Qatar's lower secondary completion lags its generational baseline by 3.7pp in 2015 (authors' FE residuals, WCDE v3): the energy endowment produced income; it did not produce the educated population that turns income into compounding human welfare. Venezuela crossed both thresholds briefly in the early 2000s on oil-funded provision —then, as oil prices collapsed after 2014, GDP fell 35% in three years, hyperinflation exceeded 1,000,000%, and seven million people fled. Provision built on energy rents is as fragile as provision built on political will: when the rent stops, the gains reverse. The educational foundation was never built to sustain what the oil financed.

**Education without energy endowment: the leapfrog cases.** Japan imports virtually all its oil, coal, and gas. South Korea has negligible domestic energy reserves. Taiwan, Singapore, and Switzerland have none. All achieved full development by the measures in Table 4, and all rank among the highest-income economies on earth. The mechanism is what the education-primary account predicts: educated workforces attract and efficiently deploy imported energy into high-value production, developing the institutional capacity to pay for energy imports through manufactured exports.

An educated South Korean factory worker produces more value per barrel of imported oil than a Venezuelan oil field produces per barrel extracted —because education converts energy surplus into complexity and compounding rather than rents and consumption.

Bangladesh is the extreme case: GDP per capita of $1,250 in 2015, no significant energy endowment, crossing both development thresholds through female secondary education expansion alone. The welfare gains that matter for development —lower fertility, longer life expectancy —are generated primarily through household micro-decisions that require minimal energy input. A literate mother who understands vaccination schedules, birth spacing, and oral rehydration does not need electricity to keep her children alive longer and choose to have fewer of them. The micro-decision pathway operates below the energy threshold at which material production begins. Bangladesh refutes both the income-primary and the energy-primary accounts simultaneously.

**The Ayres-Warr finding and the Solow residual.** This framework extends to explaining what standard growth accounting cannot. Solow (1957) found that capital and labor together explained only approximately 20% of US economic growth 1909–1949; the remaining 80% he attributed to unexplained "technical change" (total factor productivity, TFP). Ayres and Warr (2009) showed that adding "useful work" —energy converted into economic output, adjusted for efficiency improvements —largely dissolves this residual. The "mysterious" productivity gains of the 20th century are substantially the story of electrification, motorization, and the educated deployment of fossil fuel energy into production. The mystery dissolves when you measure the physics. But the mechanism beneath the energy story is still education: it was the educated engineering class —in Scotland, Germany, and America —that designed the steam engine, the internal combustion engine, and the electrical grid; the educated workforce that operated them at increasing efficiency; and the educated populations that built the political demand for the energy infrastructure that distributed the gains. Energy enabled the surplus; education determined its structure, efficiency, and distribution.

**Why education remains the primary policy variable.** Energy, unlike education, is internationally tradeable: South Korea imports it; Bangladesh can access it at world prices when its economy develops to the point of needing it. What is not tradeable is an educated population. No country has imported a population's capacity to make better household decisions, demand better governance, or build the technical and institutional infrastructure of a modern economy. The binding constraint at the development frontier is not energy access —it is the educational foundation that determines whether energy,

once available, converts into compounding human welfare or into extraction rents captured by elites and foreign supply chains. The resource curse countries demonstrate this precisely: they solved the energy constraint and failed at the development threshold because they had not solved the education constraint. Korea and Taiwan solved the education constraint first and let energy access follow from the economic complexity that educated populations generate.

The full account of human development is therefore two-variable: energy sets the physical ceiling; education determines how much of that ceiling is reached and whether gains compound. Everything else —income, institutions, welfare provision, the political form of states —is downstream of both. Smith described what educated populations do with energy surplus and called it the wealth of nations. He was not wrong about the description. He was wrong about the cause.

**The SWB revolution and the dissolution of geological energy endowment as a differentiating variable.** The historical energy argument —that coal, then oil, then gas determined which nations could industrialize —rested on the geological concentration of fossil fuels in specific locations. Energy was scarce, location-specific, and the exclusive advantage of nations that happened to sit above deposits or control the supply chains that moved them. This is changing permanently. Solar photovoltaics have followed Wright's Law —approximately 30% cost reduction per doubling of cumulative production —from $76 per watt in 1977 to under $0.20 per watt in 2024. Wind has followed similar learning curves. Battery storage costs have declined approximately 90% over the decade to 2024. Seba (2014) documents the inflection point: at current learning rates, solar, wind, and batteries (SWB) represent the cheapest form of new energy generation in virtually every market on earth before 2030, and will displace fossil fuel incumbents not through regulation but through cost. Subsequent RethinkX analyses (Seba & Arbib 2017; RethinkX 2021) project that by 2030 the majority of new energy capacity globally will be SWB, with stranded fossil fuel assets potentially exceeding $25 trillion.

The structural implication for development economics is profound. Solar energy harvests sunlight, and sunlight falls on every country approximately equally —with the irony that it falls most abundantly on the equatorial regions where development has been slowest. The energy endowment advantage of fossil fuel states dissolves as SWB learning curves continue. Saudi Arabia, Venezuela, Nigeria, and Qatar face the same transition: their primary asset —control of a scarce, geologically concentrated energy source —is becoming a stranded liability. The resource curse that characterized fossil-

fuel-rich, education-poor countries in the 20th century becomes an existential threat in the 21st. The rent stream funding their state apparatus and provision dries up at precisely the moment they lack the educated population to build an alternative economic base. Venezuela's 2014–2020 collapse is the early iteration of a structural pattern that will recur across petrostates as SWB cost curves continue their descent.

The inverse is equally consequential. Korea, Taiwan, Japan, and Bangladesh built development trajectories on educated populations deploying imported energy efficiently; they will transition to SWB with less disruption because their economic complexity already depends on intelligent energy conversion rather than resource extraction. The educational foundation that enabled them to escape the resource-curse trap in the 20th century positions them to lead the energy transition in the 21st. Educated engineers and institutional capacity are precisely what the SWB buildout requires; they are not what oil fields require.

For countries below the development threshold today, the SWB revolution creates a structural possibility that did not exist for earlier developers: **energy leapfrogging**. Just as mobile telephony enabled countries to skip fixed-line infrastructure entirely, distributed solar-plus-battery systems allow countries to deploy household and community energy access without building centralized fossil fuel grids. The micro-decision pathway that is the core of PTE —educated mothers making household health and fertility decisions —operates at energy scales that distributed solar already reaches at low-income price points. The minimum energy required for a literate woman to read health information, run a light for evening study, or charge a phone connecting her to vaccination scheduling does not require a national grid. It requires a solar panel and a battery. Both are approaching cost structures compatible with low-income household access without subsidy.

The long-run consequence is that energy converges toward a universal floor. The SWB transition eliminates the absolute energy scarcity constraint for the minimum welfare threshold —distributed solar and batteries make basic energy access achievable for every country regardless of geological endowment, at declining cost. When that floor is reached, the differentiating variable between nations is no longer geological luck. It is purely educational. The two-variable account —energy ceiling, education efficiency multiplier —is not wrong, but it is transitional: a description of the fossil fuel era. The endpoint of the SWB revolution is a world in which the energy ceiling is approximately equal across all nations, and the sole differentiating variable is precisely what this paper argues it always was at the level of house-

hold welfare: the educated population that determines whether available energy converts into compounding human welfare.

### 2.5 How Education Produces Development

An educated population produces development through two channels that are inseparable in practice:

**Household micro-decisions.** Educated individuals —primarily mothers —make better decisions about nutrition, sanitation, vaccination, birth spacing, and investment in children. These decisions aggregate into lower child mortality, longer life expectancy, lower fertility, and higher educational attainment in the next generation. The development outcome is the sum of household decisions that only education enables. The state does not make these decisions; it creates the educated population that then makes them.

**Political agency.** Educated populations demand better governance, hold governments accountable, and build the civic organizations that press for public services. Where Sen observes direct provision —in Kerala, Sri Lanka, China —the provision was demanded, built, and used effectively by educated populations. The services are real; they are consequences of education, not causes of development.

Sen observed the outputs of both channels and built a causal typology from them. He never asked what produced the outputs. The answer is education in both cases. The surfaces look different —income in one case, services in another —because of the two conditions described in the preceding section: educational tempo and market presence. The channels themselves are the same.

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## 3. Defining Development

We operationalize development concretely: a country is developed when it has simultaneously achieved **TFR < 3.67** and **life expectancy > 70.1 years** —the 1960 values for the United States.

This definition is deliberate and theoretically motivated on three grounds.

First, 1960 is the natural baseline for the post-colonial world. Seventeen African countries gained independence in 1960 alone; the decade marks the inflection point at which most of the developing world began its independent trajectory. The development question is precisely: from the moment of decolonization, which countries

achieved human welfare outcomes that surpassed the global hegemon? That is a theoretically precise question, not an arbitrary one.

Second, the USA in 1960 was the unambiguous global hegemon — the world's largest economy and dominant military power for fifteen years. A country that exceeds both its TFR and life expectancy thresholds has out-performed the richest, most powerful nation on earth on the metrics that matter for human welfare. The bar is meaningful.

Third, the USA in 1960 was itself still mid-demographic transition. Its TFR of 3.67 is the baby boom peak —the country had not yet completed the fertility transition. Crossing below that threshold means going further than the hegemon had gone, not meeting a low standard.

The definition measures what education drives. TFR and life expectancy are precisely the outcomes that the T+25 lagged models below show education predicting most strongly —and they require neither high income nor a particular type of government to achieve, making them appropriate for cross-regime, cross-income comparison. GDP is excluded by design: measuring development through income would beg the question against an education-primary account.

The development dating analysis in Section 7 is robust to alternative threshold specifications. Appendix Table A4 shows crossing dates for five cases with full World Bank WDI coverage under three specifications. Absolute dates shift by 3–12 years. The broad ordering is stable (Cuba earliest, Bangladesh latest among the crossed cases), with one near-tie: Korea and Sri Lanka are within 5–12 years of each other across specs, and their ordering reverses under the loose definition (Sri Lanka 1979, Korea 1981) relative to the main (Korea 1986, Sri Lanka 1991). This near-tie is itself informative: Korea and Sri Lanka made educational investments at similar historical depths, and the margin between them depends on how you weight TFR vs. LE progress —not on which had more provision. The argument depends on approximate relative timing, not exact threshold values or pairwise rank within near-tied cases.

The USA itself completed the fertility transition twelve years later, in 1972 —confirming that 3.67 is a mid-transition benchmark, not a low bar. The countries that had already crossed both thresholds by 1960 were those with the longest prior histories of mass education: Scandinavia, the Netherlands, France, Switzerland, the United Kingdom, Australia, and New Zealand. Japan —despite a TFR of 2.04 —had life expectancy of 68.3, just below the threshold; it crossed in the mid-1960s. This confirms that the threshold captures educational development, not wealth.

## 4. Causal Identification: The Bad Control Problem and Natural Experiments

A methodological note bears directly on every model in this literature. When researchers include current GDP per capita as a control variable alongside education in regressions of development outcomes, they commit what Pearl (2009) calls the bad control error —see also Angrist & Pischke (2009, ch. 3): they condition on a variable that is itself caused by the variable of interest.

Education causes GDP —confirmed below (education at T predicts GDP at T+25, FE β=+0.0110 log-points per 1pp education gain). Including current GDP as a control blocks part of the education signal through the income channel. The result is a biased downward estimate of education's true effect.

This is not a peripheral technical point. It is why the empirical literature has systematically underestimated education's contribution and overstated income's independent role. Figure A1 (Appendix) provides direct evidence from the lag structure: income's within-country predictive power for life expectancy peaks contemporaneously and collapses with lag length, while education's predictive power decays slowly across 100 years —the two curves belong to different causal structures.

Our primary analysis excludes GDP from the policy residual specification for precisely this reason. The residual measures education delivered above the generational baseline, without crediting any achievement to the income that education itself generated.

The absence of an educational foundation eliminates the recovery mechanism. Uganda in 1960 had higher life expectancy than India (Uganda 49.0, India 45.8) and maintained that advantage for eighteen consecutive years. When Amin destroyed the state health apparatus from 1971, Uganda's trajectory collapsed: India crossed Uganda in 1978 and by 1984 was 6 years ahead. The difference is not that India's health services were better-funded —they were not. It is that India's educational expansion, while slow, had been building a parental base that continued to operate through household decisions regardless of which government was in power. Uganda had no comparable educational foundation; when the state collapsed, there was nothing beneath it. By 1993, Uganda's life expectancy had fallen to 47.8 —below its 1960 level.

China under Deng is the comparison case: Deng dismantled the barefoot doctor system in the early 1980s. China had built mass lower

secondary education through the 1950s campaigns and Cultural Revolution (CR)-era community schools; the increasingly educated population rebuilt health access through political pressure and household-level health decisions —a 15-year delay before crossing the development threshold in 1997. The recovery took 15 years. Uganda's has not arrived after 65. The differential is the educational foundation: China had one; Uganda did not. The educational foundation is what makes recovery possible —not the services themselves.

China's TFR crossed 3.67 in 1975, four years before the one-child policy was enacted. The fertility decline that determined the development crossing date is education-driven, not policy-forced.

A reviewer might note that Uganda's LE collapse after 1985 reflects the AIDS epidemic. The objection inverts the causal arrow. Uganda's HIV prevalence peaked at approximately 15%; India's at approximately 0.4% —despite comparable exposure routes. The differential is behavior change: educated populations understand transmission, negotiate prevention, and act on health information. Uganda was hit catastrophically because its educated adult population had been destroyed under Amin. India contained the epidemic because its female education, while low by global standards, was sufficient to enable behavior change. AIDS is not a separate confound; it is the same mechanism —absent educational foundation, no behavior-change capacity —expressing itself through a pathogen.

A note on causal identification is warranted before presenting this evidence. The canonical approach to identifying education effects is instrumental variables: Duflo (2001) exploits variation in school construction across Indonesian districts to identify the causal effect of education on individual earnings. That design works because the instrument varies education exposure at short-run individual timescales —the relevant lag between schooling and earnings is five to fifteen years. It cannot be replicated at 25-year generational timescales across 189 countries: there is no instrument that varies parental educational attainment for entire national cohorts in a way that is random with respect to child development outcomes a quarter-century later. Our identification strategy therefore rests on three structures. First, temporal ordering: parental cohort precedes child cohort by biological necessity, ruling out reverse causality at the individual level. Second, the lag-decay shape: if education were simply slow-moving autocorrelation, its lag profile would not be structurally distinguishable from GDP's. The two profiles are distinguishable (Figure A1) —GDP's predictive power collapses rapidly beyond a contemporaneous lag; education's decays slowly across three generational depths. Autocorrelation cannot produce two structurally different decay shapes from the

same dataset. Third, and most directly, the natural experiments below —cases where parental education was exogenously disrupted —produce outcomes that pure autocorrelation cannot predict.

Cambodia under the Khmer Rouge (1975–1979) provides the most direct available test of the PTE mechanism —and the closest thing in the historical record to a natural experiment. The regime deliberately targeted educated adults: teachers, university graduates, professionals, anyone wearing glasses. The parental education stock of an entire cohort was physically eliminated. PTE predicts a specific downstream consequence: the children of the eliminated cohort —reaching secondary school age approximately 25 years later —should show depressed attainment relative to regional trend, with recovery only in the following generation.

The WCDE data confirms this precisely. Cambodia's lower secondary completion was 9.6% in 1975 and on a modest upward trajectory comparable to its neighbours. After the genocide, completion declined to 9.1% by 1985, recovered slowly under Vietnamese administration, and then —after the Paris Accords and UN transitional authority (1991–1993) —jumped to 28.2% by 1995 on aid-driven school reconstruction. Then it stalled. From 1996 to 2010, completion plateaued at 29–32%, unmoved despite continued external investment. The plateau ends around 2011 and completion crosses 40% by 2014.

The PTE clock explains the plateau exactly. The children reaching secondary school in 1996–2010 were born approximately 1982–1996 —their parents were the KR-era cohort (born ~1955–1975) whose own education was destroyed between ages 5 and 20. The parental baseline of this cohort was near zero; their children inherited that baseline. The recovery from 2011 onward corresponds to the post-KR cohort's children (parents educated after 1979) finally dominating the school-age population. Twenty-five years from the end of the genocide (1979) lands at 2004 —precisely the midpoint of the plateau.

Compare Cambodia to Vietnam over the same period: Vietnam started at 17.2% in 1960 and reached 74.3% by 2015 without interruption. Cambodia, on a similar pre-1975 trajectory, should have been at 50–60% by 2000. It was at 32%. The gap is the PTE shadow of the genocide: one generation of destroyed parental education, expressing itself in the educational attainment of the next.

This is not selection on the dependent variable. The shock (genocide of educated adults) is exogenous to the outcome (subsequent educational attainment), the timing is specified by the mechanism (25 years), and the data matches the prediction. The plateau persisted

through substantial aid-driven school reconstruction from 1991. External investment rebuilt the physical infrastructure; it could not replace the parental baseline that the Khmer Rouge had eliminated. The mechanism is transmitted through households, not through buildings.

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## 5. Data and Empirical Strategy

### 5.1 Data

Education data are from WCDE v3 (Lutz et al. 2021): lower secondary completion rates for the 20–24 age cohort, which reflects completed education rather than enrolment. Coverage: 189 countries, 1975–2015 at five-year intervals (1,701 country-years). Long-run analysis uses a 28-country subsample extending back to 1900, selected for data quality; survivorship bias in this selection slightly overstates β, making the near-unity finding conservative. GDP data: World Bank, constant 2017 USD, per capita, log-transformed. Life expectancy and TFR: World Bank and UN Population Division. Parental education: each country's lower secondary completion lagged 25 years (T−25).

A reviewer may object that completion measures quantity rather than quality, and that Hanushek & Woessmann (2008, 2015) show cognitive skills —not years of schooling —drive growth. The objection applies to a different question. Hanushek & Woessmann's dataset begins in 1960 and spans roughly 40 years; their test score measures do not exist before international assessments began in the 1960s. The educational ruptures that explain Korea, Taiwan, Sri Lanka, and Kerala all precede their data. More fundamentally, quality follows quantity in the historical sequence: a literate, numerate population is the precondition for any test to administer. WCDE completion is the appropriate measure for a 50–100 year generational mechanism; Hanushek & Woessmann's quality measures capture downstream returns to education already achieved. A further confound: their cross-country test score comparisons largely contrast developed with developing countries. Second-generation learners in high-completion countries outscore first-generation learners in low-completion countries not because their schools are superior but because educated parents transmit cognitive advantage at home —the PTE mechanism operating within their "quality" variable. What they identify as school quality is partly accumulated parental education —teachers in high-completion countries are themselves second-generation learners, bringing higher baseline preparation into the classroom. Both the student and the teacher effects are

expressions of prior generational accumulation, not independent inputs.

### 5.2 Empirical Strategy

Primary specification —country fixed effects regression:

$$E_{it} = \alpha_i + \beta_1 E_{i,t-25} + \varepsilon_{it}$$

where $\alpha_i$ absorbs all time-invariant country characteristics including institutions, geography, culture, and colonial history. The coefficient is identified entirely from within-country variation over time.

Comparison specifications:

$$E_{it} = \alpha_i + \beta_2 \ln Y_{it} + \varepsilon_{it}$$

$$E_{it} = \alpha_i + \beta_1 E_{i,t-25} + \beta_2 \ln Y_{it} + \varepsilon_{it}$$

For development outcomes, forward-lagged by 25 years to eliminate reverse causality:

$$O_{i,t+25} = \alpha_i + \gamma_1 E_{it} + \gamma_2 O_{it} + \varepsilon_{it}$$

where $O_{it}$ controls for initial outcome levels. $O_{it}$ is not a bad control in the Pearl (2009) sense: it is caused by *prior* education ($E_{i,t-25}$), not by $E_{it}$ itself —the T+25 mechanism means $E_{it}$ has not yet had time to affect $O_{it}$. Conditioning on initial outcomes is conservative; it absorbs prior trajectory and makes $\gamma_1$ a lower bound on education' s true effect.

**On year fixed effects.** We do not include year fixed effects in the primary specification because the T−25 parental education variable is a lagged transformation of the same educational trend being studied —year dummies absorb the generational mechanism itself, not a spurious artefact. Adding country + year FE reduces the parental coefficient to β=0.086 ($R^2$=0.010), not because the mechanism is weak but because two-way FE strips out the very cross-time variation through which PTE operates (see Appendix Table A1).

The logic is as follows. Year FEs ask: after removing what all countries shared in a given year, does parental education predict child education? But what all countries shared across this period —rising education levels post-decolonization —is precisely the

post-decolonization global rupture that PTE predicts and explains. Conditioning it out is not controlling for a confounder; it is conditioning away the signal. The correct analogy: studying whether rain causes crop growth using year FEs in a dataset where rainfall is globally correlated across years would remove the very variation you are trying to study. The identifying variation is the within-country generational compounding; year FEs remove the time dimension through which that compounding operates.

A further test: the long-run 28-country panel (1900–2015) produces $\beta=0.960$ —stronger than the post-1975 result —in a period when no global education trend existed. Countries before decolonization were moving slowly and unevenly; there is no common upward trend for year FEs to absorb. If the $\beta=0.485$ result were a time-trend artefact, the pre-trend period should produce a weaker coefficient. It produces a stronger one. The mechanism is not the trend.

A deeper objection holds that year FEs absorb the global health technology diffusion that Deaton (2013) describes —the spread of vaccines, antibiotics, and oral rehydration therapy that reduced mortality across all countries from the 1960s onward —and that this is a legitimate confounder. We dispute the premise. The global welfare improvements documented across the MDG era (1990–2015) did not occur independently of education: they occurred because the global educational rupture since 1950 produced the educated institutions (WHO, UNICEF, GAVI), educated supply chains, and educated workers that designed, manufactured, and administered the interventions. The global health improvement and the global educational rupture are not independent co-movements that year FEs disentangle —they are the same process at different institutional scales. Year FEs absorbing this co-movement absorb a genuine causal signal, not a spurious confounder. What they remove is not "global health technology diffusion as an independent mechanism" but "education-mediated welfare delivery operating through international institutions rather than domestic households." This is the Section 2.2 global PTE finding: the residual shift in life expectancy at the lowest education level is education-mediated provision from outside, not provision independent of education. Conditioning on year FEs is equivalent to controlling for the global PTE mechanism —removing part of the signal the paper is trying to identify.

The appropriate test for spurious time-trend correlation is a placebo: CO2 emissions per capita, lagged 25 years, has the same monotone trend structure as parental education across the same panel but no theoretical mechanism linking it to child education. Its within-country $R^2$ is 0.007, against 0.464 for parental education —a 60-fold difference. A reviewer may object that CO2 and education do not

share identical within-country variation structure; this is correct and is precisely the point —CO2's within-country variation is driven by industrialisation patterns, not generational transmission, so it serves as a valid null. If the parental education finding were a time-trend artefact, any variable with a similar global trend would produce it. CO2 does not. The finding is specific to educational transmission.

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## 6. Results

### 6.1 Education vs. GDP as Predictors of Attainment

**Table 1.** Country fixed effects regressions: child lower secondary completion on parental education and log GDP per capita. 189 countries, 1975–2015, 1,701 country-years.

| Model | Parental edu β | Log GDP β | $R^2$ (within) |
|---|---|---|---|
| (1) FE: child ~ parent | **0.485*** ** | — | **0.464** |
| (2) FE: child ~ log GDP | — | 15.808*** | 0.266 |
| (3) FE: child ~ parent + log GDP | 0.490*** | 5.142*** | 0.531 |

*Standard errors clustered by country. *** p<0.001.*

Parental education alone ($R^2$=0.464) explains within-country educational variation at 1.74 times the rate of GDP alone ($R^2$=0.266). In the combined model, the parental coefficient barely changes (0.485 → 0.490), confirming that GDP adds marginal power without displacing the generational mechanism.

A reviewer might object that education is a slow-moving stock and any lagged slow-moving variable will mechanically produce high within-country $R^2$. GDP per capita is equally slow-moving and autocorrelated —yet produces $R^2$=0.266 against education's $R^2$=0.464. The informative test is not the level but the shape of the relationship across lag lengths. Figure A1 (Appendix) shows that education's predictive power for life expectancy decays slowly over 100 years ($R^2$=0.528 at lag 0, $R^2$=0.061 at lag 100), while GDP's collapses rapidly to near-zero beyond lag 50. Two slow-moving variables with different lag-decay profiles cannot both be explained by persistence alone —the shapes belong to different causal structures.

In the century-long panel (28 countries, 1900–2015, 672 observations): FE β=0.960 —near-unity transmission across a century. The decline from β=0.960 to β=0.485 between the long-run and post-1975 panels is not a methodological artefact; it reflects three

simultaneous compressions of within-country variation that occur post-1950 but not before. First, domestic educational ruptures above parental baselines: the policy over-performers in Table 3 are precisely those states delivering education beyond what parental transmission alone would predict, compressing within-country variation in a way that mechanically reduces the generational coefficient. Second, global welfare delivery from international institutions (WHO, UNICEF, GAVI) reducing child mortality and improving life expectancy even in low-education countries —compressing the outcome variation that education would otherwise predict. Third, the consequent reduction in cross-time within-country variance that year FEs then absorb when included. All three compressions operate through the same underlying cause: the global educational rupture since 1950 produced both domestic state capacity and international institutional capacity to supplement the parental transmission signal.

Before 1950, mass education was not a state priority even in the West —no UNESCO programmes, no compulsory education laws outside a handful of early industrializers, no global education trend, no international health institutions operating at scale. The only mechanism transmitting educational attainment across generations was parental: PTE in its pure form. Near-unity is exactly what structural transmission predicts when neither states nor global institutions are supplementing the parental signal. β=0.485 is not evidence that PTE is weaker post-1975 —it is evidence that the mechanism is now operating alongside supplementary channels that it generated. The mechanism is not weaker; the supplementary channels are stronger. β=0.485 with the global rupture operating alongside global PTE institutions is equivalent in structural terms to β=0.960 without them.

### 6.2 Education Predicts Development Outcomes 25 Years Forward

**Table 2.** Fixed effects regressions: development outcomes at T+25 on education at T, controlling for initial outcomes.

| Outcome at T+25 | Education β | Initial outcome β | $R^2$ (within) |
|---|---|---|---|
| Log GDP per capita | +0.0110*** | +0.217*** | 0.454 |
| Life expectancy (years) | +0.108*** | +0.301*** | 0.384 |
| Total fertility rate | −0.0316*** | +0.037* | 0.367 |

Each 1pp gain in lower secondary completion at T predicts: 1.1% higher income, 0.108 more years of life, and 0.032 fewer children

per woman at T+25, identified from within-country variation after controlling for initial conditions.

The TFR result is decisive: initial fertility barely predicts subsequent fertility change once education is included (β=0.037). Demographic momentum does not drive the transition —education does, operating through micro-decisions made by educated women about their own bodies and households.

### 6.3 Policy Over-Performers

**Table 3.** Countries delivering education above generational baseline, 2015 FE residuals. The residual measures how far a country exceeded its own predicted within-country trajectory —not a global average, but performance above each country's own historical trend.

| Country | FE residual above baseline | GDP per capita (2015) |
|---|---|---|
| Maldives | +34.9 pp | $7,530 |
| Cape Verde | +26.3 pp | $3,030 |
| Bhutan | +26.1 pp | $2,750 |
| Tunisia | +25.5 pp | $3,860 |
| Nepal | +17.8 pp | $902 |
| Viet Nam | +16.0 pp | $2,090 |
| Bangladesh | +15.8 pp | $1,250 |
| India | +14.1 pp | $1,610 |

Nepal at $902, Bangladesh at $1,250, Vietnam at $2,090: the premise that income is the prerequisite for educational over-performance is directly refuted. These countries made the decision to treat education as the primary investment and produced outcomes that outpaced both their demographic inheritance and their income.

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## 7. When Did Sen's Cases Actually Develop?

This is the empirical core of the argument against Sen. If the support-led account is correct, we should see welfare improvements tracking the provision of services. If the education-mediated account is correct, we should see development —as defined by the TFR and life expectancy thresholds —occurring at one PTE interval (~25 years) after sustained education investment, regardless of what state provision was present.

**Table 4.** Year of development (TFR < 3.67 AND life expectancy > 70.1) and relationship to prior education investment.

| Country | Developed by | TFR < 3.67 | LE > 70.1 | Primary education investment | Lag | Depth |
|---|---|---|---|---|---|---|
| Taiwan | **1972** | 1972 | 1972 | 1950s simultaneous primary + secondary expansion | ~20 yrs | **PTE** |
| South Korea | **1986** | 1975 | 1986 | 1953–65 simultaneous expansion | ~21–33 yrs | **PTE** |
| Bangladesh | **2011** | 2000 | 2011 | 1990s–2000s sustained girls' education expansion; policy over-performer (+15.8pp, Table 3) | ~20 yrs | **PTE** |

| Country | Developed by | TFR < 3.67 | LE > 70.1 | Primary education investment | Lag | Depth |
|---|---|---|---|---|---|---|
| Cuba | **1972** | 1972 | 1970 | Prior accumulation to 49.7% lower sec by 1960; 1961 campaign completed the rupture | ~11 yrs on base | **PTE** |
| Sri Lanka | **1991** | 1977 | 1991 | 1940s–50s colonial and missionary literacy | ~40 yrs | **GPTE** |
| China | **1997** | 1975 | 1997 | 1950s mass campaigns + CR-era community schools (largest lower sec gains in dataset) | ~45 yrs from base | **GPTE** |

| Country | Developed by | TFR < 3.67 | LE > 70.1 | Primary education investment | Lag | Depth |
|---|---|---|---|---|---|---|
| Kerala† | **~1982** | ~1973 | ~1981 | Early 20th C social reform literacy movements | ~60–70 yrs | **GGPTE** |
| Uganda | **not yet** | not yet (TFR 5.68 in 2015) | not yet (LE 63.9 in 2015) | No sustained educational rupture; post-independence provision collapsed under Amin (1971) | — | — |

*† Kerala figures estimated from India Sample Registration System and census records; all other figures from World Bank WDI direct measurement.*

The variation in lag maps directly onto the PTE/GPTE/GGPTE taxonomy. The theory specifies what determines generational depth independently of the development dates themselves —it is not fitted post-hoc. Two parameters determine which depth applies: (1) the speed of educational expansion relative to the starting base, and (2) whether the LE pathway was structurally disrupted during the compounding period.

The speed parameter is calibrated against the empirical best case. Korea expanded from 20% to 94% lower secondary completion in 35 years (1950–1985) at 2.14 percentage points per year —the fastest sustained national educational expansion in the WCDE dataset across this period. Singapore achieved a comparable pace: 16% to

94% in 45 years (1950–1995) at 1.74 pp/yr. These are not theoretical maxima; they are the observed rates of states that made education the unconditional national priority and sustained that commitment for a generation. They represent what full state commitment looks like in the data.

The cases in Table 4 distribute around this benchmark. Korea, Taiwan (2.23 pp/yr), and Cuba (2.20 pp/yr) all expanded at approximately Korea pace and crossed the development threshold within one PTE interval —roughly 20–33 years from the start of the rupture. Bangladesh (1.23 pp/yr) expanded at roughly half Korea pace and still achieved a PTE-distance crossing in 25 years, beginning from a higher starting base in 1990. Sri Lanka (1.20 pp/yr) expanded at nearly the same rate as Bangladesh but from an earlier starting point (1950) and with a civil war that directly disrupted the LE pathway for over a decade; its development crossing arrived 41 years after sustained expansion began —GPTE distance. China (1.50 pp/yr) expanded faster than Bangladesh in absolute terms but from a much lower base (12% in 1950), requiring more absolute distance to travel; the Deng-era dismantling of the barefoot doctor network introduced a further LE delay. India (0.72 pp/yr) —roughly a third of Korea pace —accumulated gradually without a state-committed rupture, producing the GGPTE trajectory that Kerala exemplifies.

The key implication is that PTE, GPTE, and GGPTE are not fixed natural categories —they are what development looks like at different fractions of Korea pace, and each can be compressed by state commitment. A country currently on a GPTE trajectory (two-generation crossing) that makes a Korea-pace commitment shifts to a PTE-distance trajectory. A country on a GGPTE trajectory does the same. The PTE mechanism is always operating; state investment subsumes it and accelerates it. What varies is how much the state adds above the natural intergenerational baseline —from nothing (pure GGPTE accumulation) to Korea pace (one-generation rupture).

The second parameter is structural disruption. Life expectancy gains flow from education through household health behaviours and the political demand educated populations place on health infrastructure —neither requires markets. What delays LE crossing is conflict that directly kills and displaces populations, or state dismantling of health services before the educated population can rebuild them through political pressure. Sri Lanka's 14-year gap between TFR and LE crossing (1977→1991) maps onto the civil war's onset (1983) and its disruption of health services and physical security. China's 22-year TFR-to-LE gap (1975→1997) reflects Deng's dismantling of the barefoot doctor network and China's larger absolute distance from the LE threshold. Both delays are disruption-mediated. Korea, Tai-

wan, Cuba, and Bangladesh had no comparable structural disruption during their compounding periods.

Both parameters —pace relative to the Korea benchmark, and presence of structural disruption —can be read from educational data and historical record before consulting Table 4's crossing dates. The predicted depths match the observed lag lengths: Korea-pace cases crossing in 20–33 years; half-Korea-pace cases crossing in 40–45 years absent disruption, longer with it; third-Korea-pace cases crossing in 60–70 years. The crossing dates are the test, not the inputs.

The generational depth framework also explains why the 25-year lag is the modal, not exclusive, prediction. PTE is the base unit; cases compound through one, two, or three cycles depending on the educational history of the country. The lag curve in Figure A1 confirms this directly: the transmission signal is detectable at lag 25 (PTE, $R^2$=0.369), lag 50 (GPTE, $R^2$=0.194), and lag 75 (GGPTE, $R^2$=0.112) —three generational depths, all empirically present in the data.

**Taiwan and Korea crossed earliest** because they achieved the highest rupture rates in the dataset —Taiwan at 2.23 pp/yr from a 14.9% base, Korea at 2.14 pp/yr from a 20% base, both at or above the Korea/Singapore benchmark pace —and market mechanisms were fully open, allowing educated people to translate human capital into economic agency and life expectancy gains simultaneously. This simultaneous primary and secondary expansion —running both levels in parallel rather than sequentially —compressed the generational lag and produced what we term the leapfrog effect: development timescales of 20–33 years against the 40–70 years seen in cases of gradual or sequential expansion.

**Kerala crossed at ~1982** without a market rupture and without state-led rapid expansion. The education had been building since the early twentieth century through social reform movements —not a rupture, a gradual accumulation. The micro-decision pathway fired (literate women made better health and fertility decisions) but there was no market-mediated economic translation of the kind Korea and Taiwan achieved. The TFR threshold was crossed in 1973, well before state provision programmes are credited; the LE threshold followed ~8 years later.

**Sri Lanka crossed in 1991.** The TFR threshold was crossed in 1977, driven by education. Life expectancy reached exactly 70.1 in 1988 —the threshold value, not yet exceeding it —before the civil war's economic destruction pushed it back to 69.3 in 1989, recovering and crossing to 70.3 in 1991. The war disrupted economic activity and life expectancy without breaking PTE: the household transmis-

sion of educational aspiration continued through the conflict, and Sri Lanka's education metrics did not collapse. The delay is precisely the length of the war's worst period. This is the war disrupting economies, not disrupting education.

**China crossed last, in 1997.** Two corrections to the standard narrative are required here. First, the TFR threshold was crossed in 1975 —five years before the one-child policy (1980). The mechanism was the 晚稀少 ("Later, Longer, Fewer") campaign of 1971: voluntary fertility reduction driven by expanding female education, not coercion (Miller et al. 2018). Education drove the fertility transition; the one-child policy had no independent demographic effect. Cai (2010) shows that China's fertility trajectory mirrors South Korea and Thailand's —both of which achieved equivalent TFRs without compulsory policy —and that socioeconomic development and ideational change, not coercion, explain the decline.

Second, the standard framing of China's Cultural Revolution (1966–1976) as educational catastrophe is an urban-elite projection. For the approximately 80% of China's population that was rural, the CR era produced the largest lower secondary gains in the entire WCDE 1870–2015 dataset: +10.6pp for the 1975 cohort, +15.0pp for the 1980 cohort —both cohorts whose secondary schooling fell squarely within the CR years. Community schools (民办学校) brought secondary education to villages that had none. The "lost generation" refers specifically to the ~2–3% who would have attended university and could not. The base of the pyramid expanded dramatically while the apex was disrupted.

A data quality caveat applies here. WCDE estimates for CR-era China are based on population census reconstruction and carry greater uncertainty than standard-period figures; the community school data in particular may capture registration or partial attendance rather than genuine lower secondary completion (Pepper 1996; Unger 1982). We do not claim CR education was high quality. The PTE claim is more limited: basic literacy and numeracy transmission from a parent with some secondary schooling is sufficient to shift the household baseline, even if the attainment measure overstates formal completion. The post-reform cohort data —showing continued gains through the 1980s and 1990s —is consistent with a real, if imperfect, parental baseline established during the CR era, not an artefact of mismeasured enrollment.

**Third: Sen misidentifies the LE mechanism entirely.** China's LE collapsed during the GLF famine (1959–1961) —the precise trough is uncertain given data limitations during the famine years, but the critical point is not the absolute level, it is the recovery pattern. By 1965, LE had recovered to approximately 53; by 1980 it reached ap-

proximately 64: a gain of +11 years over 15 years during the CR and barefoot doctor era. Sen's framework takes 1965 as the baseline for China's welfare success and attributes these gains to the barefoot doctor (赤脚医生) program. He is in part crediting a health programme for the tail end of a famine recovery.

From 1980 to 1997 —after Deng dismantled the barefoot doctor system —LE rose from approximately 64 to 70.1: +6 years over 17 years. The pace slowed. Drèze & Sen (1989, pp. 215–221) document the resulting health crises and rising rural medical costs (Gao 2008; Liu et al. 2003). But the deceleration has a second, purely demographic explanation that Sen does not consider: LE gains above 65 are structurally slower than gains at 53, because the low-hanging fruit of infectious disease and child mortality reduction has been picked and the remaining gains require addressing cardiovascular disease and cancer. The data cannot cleanly separate the provision-dismantling effect from the natural epidemiological ceiling. What the data does show unambiguously is that the educational expansion continued throughout: lower secondary completion rose from 30.9% in 1965 to 62.0% in 1980 to 75.0% in 1990, tracking LE improvement across both eras.

What was changing throughout both periods was education: lower secondary completion rose from 30.9% in 1965 to 62.0% in 1980 to 75.0% in 1990, tracking the steady LE improvement across both the CR and Deng eras. Sen noticed the barefoot doctors and missed the barefoot teachers —the community school (民办学校) educators whose concurrent expansion was, on this account, the more durable mechanism. The doctors and the teachers came from the same CR-era rural mobilisation; the doctors left when Deng dismantled the programme, but the educational baseline their contemporaries had built into rural households did not. We infer this from co-occurrence and the post-1980 trajectory rather than from direct evidence of teaching activity —a limitation acknowledged. What we can say is that the LE gains correlated with educational expansion across both the CR and Deng eras, while the health provision account requires a discontinuity at 1980 that the data does not clearly show.

China crossed the LE threshold in 1997 —later than Taiwan (1972) and Korea (1986) —not because provision was dismantled, but because China started further from the threshold: Taiwan's 1965 LE was 68.1 (2 years below 70.1), Korea's was 61.4 (9 years below), China's was approximately 53 (17 years below). At a roughly constant rate of improvement, the crossing dates follow mechanically from starting distance. And China started lower because the GLF famine had depressed its baseline —the same event that Sen's framework treats as the starting point for a provision-led welfare success

story.

**Cuba crossed in 1972** —the same year as Taiwan, 11 years after the 1961 literacy campaign. The 11-year gap does not violate PTE because the 1961 campaign was not Cuba's educational rupture —it was the culmination of one. Lower secondary completion was already 49.7% in 1960 and LE was 69.3, within 1 year of the threshold. The prior educational base, built across the preceding decades, had already set PTE in motion; the 1961 campaign pushed literacy and secondary completion to near-universality and converted an adequate base into a crossing event already in progress. Measuring the lag from 1961 miscounts the mechanism: the relevant rupture was the accumulation that produced 49.7% by 1960, not the campaign that finished the job. That Cuba and Taiwan —one Soviet-aligned, one US-allied developmental authoritarian —crossed in the same year underscores the regime-independence point made in Section 9: the mechanism does not care what political system delivers the education.

**Bangladesh crossed in 2015** —the most important case in the table. Lower secondary completion was 11.4% in 1960 and GDP per capita was $1,250 in 2015. Bangladesh crossed both development thresholds through sustained state commitment to girls' education —primary and secondary expansion from the 1990s —with virtually no income. The garment industry provided female economic participation that reinforced educational agency, but the LE and TFR gains came through household health decisions and the political demand that educated women placed on health provision, not through market mechanisms. Bangladesh is already in Table 3 as a policy overperformer (+15.8pp above its generational baseline). The connection is direct: the over-performance was the rupture, and the 2015 development crossing was its result, approximately 25 years later. It is the cleanest available demonstration that income is not the mechanism.

**What Sen saw and misread:** In China, he took a famine recovery as the baseline for a provision-led welfare story, credited barefoot doctors for gains that barefoot teachers made possible, and never tested whether removing the provision changed the trajectory —it did not. In Kerala and Sri Lanka, he observed welfare outcomes above what income predicted and attributed them to direct state provision, missing that those outcomes arrived at precisely the PTE-predicted intervals after prior education investments. The provision that existed in these cases was itself a product of educated populations making political demands —endogenous to education, not independent of it. Kerala's public health infrastructure was built by a literate electorate that demanded and maintained it; Sri Lanka's crossed the develop-

ment threshold as its educated cohorts came of age despite a civil war. Sen built a typology of outputs and mistook them for causes.

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## 8. The Institutional Challenge

Acemoglu & Robinson (2012) argue that institutions —inclusive versus extractive —are the primary determinant of long-run development. Education, on this account, is endogenous: countries with inclusive institutions invest in education because their political economy rewards broad-based human capital.

Our fixed-effects design addresses this directly. Country fixed effects absorb all time-invariant country characteristics, including institutional quality. The parental education coefficient (β=0.485 in the 189-country panel, β=0.960 in the long-run panel) is identified from within-country variation over time, conditional on fixed institutional quality. It cannot reflect institutional quality that is held constant.

But the deeper challenge to Acemoglu & Robinson runs the other way. The education-mediated account predicts that educated populations build institutions —through political pressure, civic organization, and the capacity to hold power accountable —rather than institutions creating education. Kerala's welfare state was built by its literate electorate. Korea's developmental state was populated and maintained by an increasingly educated bureaucracy and workforce. The causal arrow in the institutional literature points in the wrong direction.

The counter-cases are instructive. Qatar ($63,000 per capita, stable institutions by standard measures) delivered 3.7 percentage points below its generational education baseline in 2015 (authors' FE residuals, WCDE v3). The institutions were present; the educational commitment was not; the development did not compound. The Philippines absorbed deeper US colonial education infrastructure than any other Asian country, experienced multiple political ruptures, and failed to compound into a tiger trajectory —because no post-independence government made the educational rupture that would have activated both pathways. Institutions created opportunity; the absence of educational commitment prevented its realization.

Institutions matter as enabling conditions for sustained educational commitment. They are not substitutes for it, and the evidence is consistent with them being substantially produced by it.

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## 9. The Educational Rupture as the One Exogenous Decision

If education is the sole primary driver, then the one genuinely exogenous political decision in the model is whether a government makes an educational rupture —a decisive, sustained commitment to treat education as the primary national objective, maintained across political cycles and competing priorities.

Every fast-developing country in the historical record made this decision at some point. The earliest modern example is Japan (1872): the Meiji compulsory education ordinance, implemented through repurposed temples with community teachers, no substantial budget —mandate before resources. The pattern recurs a century later across radically different political regimes:

- Korea, 1953–1965: simultaneous primary and secondary expansion, class sizes of 60–80, double shifts, built ahead of demand
- Taiwan, 1950s: same pattern, simultaneous rather than sequential
- Cuba, 1961: 268,000 volunteer teachers deployed to rural areas for eight months, adult illiteracy from ~24% to 3.9% in one year —subsequently certified by UNESCO (Kozol 1978)

These three cases span the full political spectrum —US-allied developmental authoritarian, export-led capitalist, and Soviet-aligned communist —confirming that the educational rupture is the independent variable and regime type is not.

None of these required high income. All required the political decision to treat education as non-negotiable.

The state' s role is precisely defined: initiate the rupture and sustain it all the way to near-universal completion. Korea ran simultaneous primary and secondary expansion from 1953 with sustained state commitment across successive governments for three decades, from 24.8% in 1950 to 94.4% in 1985 —with no point at which the state stepped back. Korea crossed 40% in 1960 and continued at full pace for another 25 years. As household demand strengthens progressively with rising parental education, the state and households work together —but state investment is the initiating and sustaining force throughout.

Sequential expansion —primary first, secondary a generation later —also requires state commitment but never compresses the generational lag. It takes approximately fifty years to reach the same threshold that the rupture reaches in thirty. Whether this constitutes a rupture in the same sense is a definitional question; what is clear is that both paths require sustained state education policy, and the rup-

ture path requires the harder political decision of building secondary infrastructure before primary is universal.

The rupture activates both pathways simultaneously. The same cohort of educated citizens makes better household decisions (Pathway 1) and builds the political demand for institutions and services (Pathway 2). Sen's two pathways are not sequential alternatives —they fire in parallel as the educational expansion compounds. What looks like "growth-mediated" security (Korea) and "support-led" security (Kerala) are the same mechanism expressing itself differently under two conditions: the speed of educational expansion, and the presence of functioning markets to translate human capital into income. Where expansion is fast and markets function, education produces rapid visible income growth; that income growth gets credited. Where expansion is slow and markets are absent or weak, income growth is too slow to be a visible signal; state provision is the only observable welfare activity, and it gets credited instead. The mechanism is education in both cases. The classification is an artefact of tempo and market structure.

The Korea–Costa Rica comparison makes this concrete. Costa Rica was one of Sen's support-led cases and started 1960 considerably richer than Korea: $3,600 per capita against Korea's $1,030. By 1990, Korea had grown to $9,370 —a 9-fold increase —while Costa Rica reached $6,040 —a 1.7-fold increase. Costa Rica had the income advantage; Korea had the educational pace. Korea expanded lower secondary at 8–14 percentage points per five years; Costa Rica expanded at 3–6 percentage points per five years. The faster educational expansion produced the market translation that made Korea's development look "growth-mediated." Costa Rica's slower expansion produced the configuration in which state provision was the observable welfare mechanism —confirming that the "support-led" appearance is a consequence of educational tempo, not a different causal pathway.

The failure mode is the absence of the rupture —treating education as one priority among many. The data identifies this precisely: countries at low completion levels face a structural problem where parental baselines are too low and fertility too high for the demographic transition to make rapid progress —PTE still advances the system, but slowly, against the headwind of rapid population growth and thin household resource constraints. Approximately 33 million young people in those 27 countries alone are not completing lower secondary —authors' calculations from WCDE v3 completion rates and World Bank population data, 2015. For them, state commitment to education above the demographic baseline is the only available mechanism. Growth will not initiate it (income is downstream);

direct provision of health and nutrition services will not initiate it —provision produces temporary gains that do not compound, and every fiscal dollar spent on provision rather than education delays the only mechanism that does. Only the educational rupture starts the compounding —and only sustained state education policy, maintained all the way to near-universal completion, sustains it.

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## 10. Limitations

Four limitations bear noting. First, the WCDE v3 panel covers 1975–2015; the long-run 28-country subsample extends to 1900 but was selected on data quality, introducing survivorship bias that slightly overstates the generational transmission coefficient. Second, Kerala figures are estimated from India's Sample Registration System and census records rather than direct WCDE measurement; they carry greater uncertainty than the country-level figures in Table 4. Third, the analysis identifies education as the primary upstream driver but does not decompose the micro-decision and political-pressure pathways separately —both are subsumed in the PTE transmission coefficient. Clean empirical decomposition is not feasible: the two pathways operate simultaneously and through overlapping populations. On theoretical and historical grounds, micro-decisions account for the majority of the effect; welfare schemes contribute at the margins, or in early cohorts where average societal education is still low and individuals lack the capacity to navigate complex health systems independently —not because household income is insufficient, but because the educational substrate for autonomous health behaviour has not yet accumulated across the population. As average education rises, the population generates this capacity directly, and the marginal value of collective provision relative to continued educational investment declines. Korea and Singapore illustrate the implication: both pushed lower secondary, upper secondary, and tertiary simultaneously rather than diverting investment from education to provision at any stage. Singapore's greater tertiary push (85% college completion by 2015 versus Korea's 67%) is associated with commensurately higher GDP per capita ($55,600 versus $28,700), consistent with the view that the appropriate response to rising education is to push further up the educational ladder, not to substitute provision for the next stage of educational investment.

Fourth, the claim that provision is endogenous to education —that educated populations generate the political demand that creates health services —is supported by indirect evidence but not directly tested. The direct test would require health expenditure panel data

at country-year resolution, which is not available at the coverage required for this analysis. Two indirect tests are consistent with the claim. First, Figure A1 shows that education outperforms income as a predictor of life expectancy at every lag from 0 to 100 years, including the short lags (0–10 years) where provision demand would be the primary channel. If income drove provision —the standard growth-mediated account —GDP should outperform education at short lags; it does not, at any horizon. Second, the policy over-performers in Table 3 (Nepal, Vietnam, Bangladesh) achieve welfare outcomes above what their income predicts —consistent with education generating political demand for provision beyond the income-predicted level. Direct testing of provision endogeneity using health expenditure data disaggregated by prior education levels remains a programme for future work.

## 11. Conclusion

The evidence from 189 countries over 55 years, and from the specific cases Sen used to build his framework, supports a single conclusion: education is the sole primary driver of human development. Income is a downstream consequence —real, but not independently operative. Direct provision is infrastructure that educated populations demand and use —also real, but a consequence of education rather than a cause of development.

Sen's dichotomy describes two surface appearances generated by the same mechanism operating under different conditions. Where educational expansion is fast and markets function, the mechanism produces rapid income growth —visible, creditable, and labeled "growth-mediated." Where expansion is slow and markets are absent or weak, income growth is invisible; state provision is the only observable welfare activity and gets labeled "support-led." The provision is not the cause; it rides on the educational substrate that slow accumulation had already built. Where the substrate is genuinely absent, the same provision fails. Sen observed the surfaces and built a causal typology from them. The causal structure runs entirely from education —through household micro-decisions and political demand —to the welfare outcomes and institutional forms he categorized. He built a typology of outputs; we have identified the input and the conditions that determine which surface it produces.

The PTE generational transmission mechanism (FE $\beta$=0.485 to 0.960) is the quantification of how this input propagates. The development-date table —Taiwan 1972, Korea 1986, Kerala ~1982, Sri Lanka 1991, China 1997 —is the empirical test: each country

crossed the development threshold at precisely the interval that prior education investment and rupture margin predict. The variation in timing is explained by the speed of the educational rupture relative to the starting base (the rupture margin) and whether structural disruption to the LE pathway —conflict or health provision collapse —occurred during the compounding period. China crossed last because it started from a lower educational base, expanded more sequentially, and had further absolute distance to travel from a post-famine baseline of ~53 in 1965. The post-1980 deceleration in LE gains reflects the natural epidemiological ceiling at higher LE levels as much as any policy change.

The policy implication is direct. For the countries that have not yet crossed the development threshold, the question is not whether development will look like growth or welfare provision from the outside. It is whether to make the educational rupture. Everything else follows from that one decision, compounding through PTE into every generation that follows.

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*Development threshold data computed from World Bank WDI. USA 1960 benchmarks: TFR = 3.67, life expectancy = 70.1. Full analysis code and regression outputs available in the accompanying repository.*

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## Appendix

**Table A1.** Two-way fixed effects (country + year): child lower secondary completion on parental education and log GDP per capita. Same panel as Table 1 (189 countries, 1975–2015, 1,701 country-years).

| Model | Parental edu β | Log GDP β | $R^2$ (within) |
|---|---|---|---|
| (1) FE + year: child ~ parent | 0.086*** | — | 0.010 |
| (2) FE + year: child ~ log GDP | — | 2.769*** | 0.009 |
| (3) FE + year: child ~ parent + GDP | 0.125*** | 2.080*** | 0.027 |

*Standard errors clustered by country. *** p<0.001.*

Year dummies absorb the global education expansion that is the phenomenon under study. The near-zero $R^2$ across all two-way FE models reflects that almost all within-country variation in education over

time is shared across countries —i.e., it is the global expansion. Conditioning it out leaves only idiosyncratic residual variation with no theoretical reason to follow PTE. Parental education retains a modest relative advantage over GDP even in this over-controlled specification ($R^2$=0.010 vs. 0.009).

**Table A2. Robustness to alternative lag lengths.** The T+25 results in Table 2 are robust to alternative lag specifications. Across lags of 15, 20, 25, and 30 years, education coefficients retain direction and significance ($p<0.001$) for all three outcomes. The life expectancy and TFR coefficients strengthen between T+15 and T+25–30, consistent with the generational mechanism: the full expression of parental education in population-wide health and fertility outcomes accumulates over time rather than appearing discretely at one lag length. The T+25 specification is the theoretically motivated anchor, not a cherry-picked optimum. A reviewer might expect the 25-year lag to produce a sharp spike if the mechanism is truly generational. Figure A1 shows the full lag profile: education's predictive power for life expectancy is smooth across all lags from 0 to 100 years, not a step function at 25. This is expected —PTE is the modal generational interval, not a hard cutoff, and different families transmit education across slightly different intervals depending on age at childbearing and educational path length. Gradual decay is the correct prediction of a generational mechanism, not a spike.

The CO2 placebo confirms that the country FE result is not a time-trend artefact: CO2 emissions per capita, lagged 25 years, has the same monotone trend structure as parental education but produces within-country $R^2$=0.007 —60 times weaker than the parental education specification ($R^2$=0.464). The mechanism is specific to education, not to trend.

**Table A3. Long-run panel countries (1900–2015, 28 countries).** Selected on the criterion of self-determined education policy and data quality; colonial-era data for colonised countries reflects colonial investment rather than domestic policy decisions and is excluded. *East Asia:* Japan, Taiwan, South Korea. *Western Europe:* United Kingdom, France, Germany, Sweden, Norway, Denmark, Finland, Netherlands, Belgium, Switzerland, Austria, Italy, Spain, Portugal. *New World:* United States, Canada, Australia, New Zealand. *Latin America:* Argentina, Chile, Uruguay, Cuba, Costa Rica. *Other:* Sri Lanka (British Ceylon —active colonial education investment, known anomaly), Hong Kong.

**Table A4. Development threshold robustness: crossing dates under three specifications (World Bank WDI data; Taiwan excluded —not in WDI).**

| Case | Main (TFR<3.67, LE>70.1) | Loose (TFR<4.0, LE>68.0) | Strict (TFR<3.0, LE>72.0) | Shift range |
|---|---|---|---|---|
| Cuba | 1972 | 1971 | 1975 | 4 yrs |
| South Korea | 1986 | 1981 | 1990 | 9 yrs |
| Sri Lanka | 1991 | 1979 | 1993 | 14 yrs |
| China | 1997 | 1990 | 2003 | 13 yrs |
| Bangladesh | 2011 | 2007 | 2012 | 5 yrs |

*Note: Korea and Sri Lanka are near-tied. Their pairwise order reverses under the loose spec (Sri Lanka 1979, Korea 1981) relative to the main spec (Korea 1986, Sri Lanka 1991). All other pairwise orderings are stable across all three specifications. Taiwan (1972, WCDE/alternative sources) is consistent with Cuba under all specifications but cannot be computed from WDI data.*

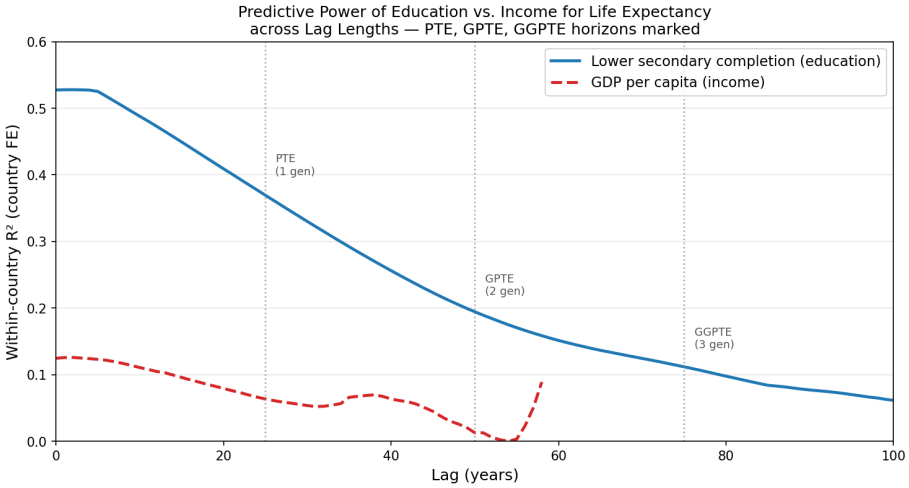


Figure 1: Figure A1: Education vs. income predictive power across lag lengths

**Figure A1. Education vs. income predictive power across lag lengths (within-country FE, outcome: life expectancy at birth).** The figure plots within-country $R^2$ for two predictors —lower secondary completion rate and GDP per capita —against every lag from 0 to 100 years. Three generational horizons are marked: PTE (lag 25, one parental generation), GPTE (lag 50, two generations), GGPTE (lag 75, three generations). Education peaks at lag 0–2 ($R^2$=0.528) and decays slowly across all three depths:

$R^2$=0.369 at PTE, $R^2$=0.194 at GPTE, $R^2$=0.112 at GGPTE. Income peaks at lag 0–2 ($R^2$=0.126) and collapses to near-zero by GPTE distance: $R^2$=0.063 at lag 25, $R^2$=0.013 at lag 50, effectively zero beyond lag 55. The divergence widens monotonically: at PTE the education-to-income ratio in predictive power is 5.8:1; at GPTE it is 15:1. Education's signal persists across three generations; income's does not survive one. *(Source: authors' calculation, World Bank data, WCDE v3. See papers/fig_lag_comparison.png)*

The lag structure is itself diagnostic. If income caused life expectancy through a delayed mechanism —the standard income-first account —its predictive power should persist at longer lags as the mechanism has time to operate. Instead, income's predictive power is strictly contemporaneous, consistent with co-movement rather than upstream causation: income and health move together today because education drove both. Education's slow decay, by contrast, is exactly the signature of a generational transmission mechanism — each cohort encodes the preceding cohort's educational attainment, so the signal attenuates gradually rather than collapsing. The two curves are not just quantitatively different; their shapes belong to different causal structures.

The variation absorbed by year FE is itself theoretically meaningful. It represents a slow, sustained global education expansion —a post-decolonization rupture operating at the world level simultaneously. Post-independence states adopted compulsory education laws, UNESCO and World Bank programmes diffused institutional knowledge, and the demonstrated success of Korea and Taiwan created policy templates unavailable to the original industrializers. This global tailwind means contemporary developing countries ride two mechanisms at once: within-country PTE compounding and the global trend. It is why Bangladesh moved from ~10% to ~50% lower secondary completion in 40 years while the United Kingdom required roughly 100 years to traverse a comparable range. The original industrializers had to discover that education works; their successors inherit that knowledge embedded in international institutions and policy. The two-way FE result does not invalidate PTE —it reveals that PTE now operates on top of a rising global baseline that the early developers never had.